Programming 18: Smart Payslip Generator (Single Employee)

Input Requirements

Use a Scanner object to collect the following inputs:

1. Employee Information

- Employee Name
- Employee ID
- Department
- Basic Monthly Salary

2. Payroll Information

- Payroll Start Date (YYYY-MM-DD)
- Payroll End Date (YYYY-MM-DD)
- Payout Date (YYYY-MM-DD)

3. Work Details

- Total Regular Hours Worked
- Overtime Hours Worked
- Number of Absent Days

4. Allowances

- Rice Allowance
- Transportation Allowance
- Communication Allowance

5. Deductions

- SSS Contribution
- PhilHealth Contribution
- PAG-IBIG Contribution
- Withholding Tax
- Other Deductions

Computation and Business Logic

Your program must perform the following:

- **Hourly Rate:**
  Basic Monthly Salary ÷ 160
- **Daily Rate:**
  Basic Monthly Salary ÷ 20
- **Regular Pay:**
  Regular Hours × Hourly Rate
- **Overtime Pay:**
  Overtime Hours × (Hourly Rate × 1.25)
- **Absence Deduction:**
  Absent Days × Daily Rate
- **Total Allowances:**
  Sum of all allowances
- **Gross Pay:**
  Regular Pay + Overtime Pay + Total Allowances
- **Total Deductions:**
  (SSS + PhilHealth + PAG-IBIG + Tax + Other Deductions + Absence Deduction)
- **Net Pay:**
  Gross Pay – Total Deductions

Output Requirements

Display a **well-formatted payslip** with the following sections:

*** Smart Employee Payslip ***

Employee Details

- Name
- Employee ID
- Department

Payroll Summary

- Pay Period
- Payout Date

Earnings

- Regular Pay
- Overtime Pay
- Total Allowances
- Gross Pay

Allowances Breakdown

- Rice Allowance
- Transportation Allowance
- Communication Allowance

Deductions

- SSS
- PhilHealth
- PAG-IBIG
- Withholding Tax
- Absence Deduction
- Other Deductions
- Total Deductions

Net Pay

✓ All monetary values must be formatted to **2 decimal places**
✓ Align output for readability

Sample Input

- Name: Maria Santos
- ID: EMP-010

- Department: IT
- Salary: 40000
- Start Date: 2025-09-01
- End Date: 2025-09-15
- Payout: 2025-09-20
- Regular Hours: 80
- Overtime Hours: 6
- Absent Days: 1
- Rice Allowance: 1500
- Transportation Allowance: 1200
- Communication Allowance: 800
- SSS: 1500
- PhilHealth: 600
- PAG-IBIG: 100
- Tax: 3000
- Other Deductions: 200

Programming 19: Student Information and Tuition Fee Billing System

Input Requirements

Use a Scanner object to collect the following data:

1. Student Information

- Student Name
- Student ID
- Course/Program (e.g., BSIT, BSCS, BSEMC)
- Year Level

2. Enrollment Details

- Number of Units Enrolled
- Tuition Fee per Unit
- Miscellaneous Fee

3. Additional Fees

- Laboratory Fee
- Library Fee
- Other Fees

4. Payment Details

- Down Payment
- Scholarship Discount (if any)

Computation and Business Logic

Your program must perform the following calculations:

- **Tuition Fee Total:**
  Number of Units × Tuition Fee per Unit
- **Total Additional Fees:**
  Laboratory Fee + Library Fee + Other Fees
- **Gross Total Fee:**
  Tuition Fee Total + Miscellaneous Fee + Total Additional Fees
- **Discount Amount:**
  Apply the Scholarship Discount directly as input value
- **Net Total Fee:**
  Gross Total Fee – Scholarship Discount
- **Balance:**
  Net Total Fee – Down Payment

Output Requirements

Display a **well-formatted billing statement** in the console:

*** STUDENT TUITION BILLING SYSTEM ***

Student Details

- Name
- Student ID
- Course
- Year Level

Enrollment Summary

- Units Enrolled
- Tuition per Unit
- Tuition Fee Total

Fees Breakdown

- Miscellaneous Fee
- Laboratory Fee
- Library Fee
- Other Fees
- Gross Total Fee

Payment Summary

- Scholarship Discount
- Net Total Fee
- Down Payment
- Remaining Balance

✓ All monetary values must be formatted to **2 decimal places**
✓ Output should be **clean and properly aligned**

Sample Input

- Name: Juan Dela Cruz
- ID: 2025-001

- Course: BSIT
- Year: 1
- Units: 21
- Fee per Unit: 1500
- Misc Fee: 5000
- Lab Fee: 2000
- Library Fee: 1000
- Other Fees: 500
- Down Payment: 10000
- Scholarship Discount: 3000

Programming 20: Customer Purchase Order with Receipt System

Input Requirements

Use a Scanner object to collect the following:

1. Customer Information

- Customer Name
- Customer ID

2. Purchase Details

The system should allow input for **at least 3 items**:

For each item:

- Item Name
- Quantity
- Price per Item

3. Payment Details

- Discount (%)
- Amount Tendered by Customer

Computation and Business Logic

Your program must perform the following:

- **Item Total (per item):**
  Quantity × Price
- **Subtotal:**
  Sum of all item totals
- **Discount Amount:**
  Subtotal × (Discount / 100)
- **Total Amount Due:**
  Subtotal − Discount Amount
- **Change:**
  Amount Tendered − Total Amount Due

Output Requirements

Display a **well-formatted receipt** in the console:

******** CUSTOMER RECEIPT ********

Customer Details

- Name
- Customer ID

Purchased Items

Display in tabular format:

- Item Name
- Quantity
- Price
- Total

Payment Summary

- Subtotal
- Discount (%)
- Discount Amount
- Total Amount Due
- Amount Tendered
- Change

✓ All monetary values must be formatted to **2 decimal places**
✓ Align columns properly for readability
✓ Use clear labels and separators

Sample Input

- Name: Maria Santos
- ID: CUST-001

Items:

1. Mouse – Qty: 2 – Price: 250
2. Keyboard – Qty: 1 – Price: 750
3. USB Drive – Qty: 3 – Price: 300

- Discount: 10%
- Amount Tendered: 2000

Programming 21: Simple Ticketing System (Plain Ticket)

Input Requirements

Use a Scanner object to collect the following:

1. Customer Information

- Customer Name
- Customer ID

2. Ticket Details

- Event Name
- Ticket Type (e.g., Regular, VIP)
- Ticket Price
- Number of Tickets

3. Additional Options

- Discount (%) (if applicable)

4. Payment Details

- Amount Tendered

Computation and Business Logic

Your program must perform the following:

- **Total Ticket Cost:**
  Ticket Price × Number of Tickets
- **Discount Amount:**
  Total Ticket Cost × (Discount / 100)
- **Final Amount Due:**
  Total Ticket Cost – Discount Amount
- **Change:**
  Amount Tendered – Final Amount Due

Output Requirements

Display a **plain formatted ticket** in the console:

******** EVENT TICKET ********

Customer Details

- Name
- Customer ID

Ticket Information

- Event Name
- Ticket Type
- Ticket Price
- Number of Tickets
- Total Ticket Cost

Payment Summary

- Discount (%)
- Discount Amount
- Final Amount Due
- Amount Tendered
- Change

✓ All monetary values must be formatted to **2 decimal places**
✓ Output should be **clean, aligned, and easy to read**
✓ Use separators (lines) to organize sections

Sample Input

- Name: Juan Dela Cruz
- ID: CUST-101
- Event: CCS Tech Expo 2026
- Ticket Type: VIP
- Ticket Price: 500
- Quantity: 3
- Discount: 10%
- Amount Tendered: 2000

Programming 22: Simple Reservation and Billing System

Input Requirements

Use a Scanner object to collect the following:

1. Customer Information

- Customer Name
- Customer ID

2. Reservation Details

- Reservation Type (e.g., Room, Cottage, Table)
- Number of Days / Hours Reserved
- Rate per Day / Hour

3. Additional Services

- Food Charges
- Service Charges
- Other Charges

4. Payment Details

- Discount (%)
- Down Payment

Computation and Business Logic

Your program must perform the following:

- **Reservation Cost:**
  Number of Days/Hours × Rate per Day/Hour
- **Total Additional Charges:**
  Food Charges + Service Charges + Other Charges
- **Subtotal:**
  Reservation Cost + Total Additional Charges

- **Discount Amount:**
  Subtotal × (Discount / 100)
- **Total Bill:**
  Subtotal – Discount Amount
- **Remaining Balance:**
  Total Bill – Down Payment

Output Requirements

Display a **well-formatted reservation bill** in the console:

******** RESERVATION BILLING ********

Customer Details

- Name
- Customer ID

Reservation Details

- Reservation Type
- Duration (Days/Hours)
- Rate
- Reservation Cost

Charges Breakdown

- Food Charges
- Service Charges
- Other Charges
- Subtotal

Payment Summary

- Discount (%)
- Discount Amount
- Total Bill

- Down Payment
- Remaining Balance

✓ All monetary values must be formatted to **2 decimal places**
✓ Output should be **clear, aligned, and easy to read**
✓ Use separators for better readability

Sample Input

- Name: Maria Santos
- ID: RES-001
- Reservation Type: Room
- Days: 2
- Rate per Day: 2500
- Food Charges: 1200
- Service Charges: 500
- Other Charges: 300
- Discount: 10%
- Down Payment: 2000

PROGRAMMING 23 **Simple CRS – Car Rental System**

You are required to develop a **Java console application** that simulates a **Car Rental System (CRS)**. The program will allow a user to input customer and vehicle rental details, compute the total rental cost, and generate a **formatted rental receipt**.

This system represents a simplified version of real-world **vehicle rental services**, handling **rentals, reservations, and payments**.

Input Requirements

Use a Scanner object to collect the following:

1. Customer Information

- Customer Name
- Customer ID
- Driver's License Number

2. Vehicle Details

- Vehicle Type (e.g., Sedan, SUV, Van)
- Vehicle Model
- Rental Rate per Day

3. Rental Details

- Number of Rental Days
- Pickup Date (YYYY-MM-DD)
- Return Date (YYYY-MM-DD)

4. Additional Charges

- Insurance Fee
- Fuel Charge
- Other Charges

5. Payment Details

- Discount (%)
- Amount Paid

Computation and Business Logic

Your program must perform the following:

- **Rental Cost:**
  Rental Rate per Day × Number of Rental Days
- **Total Additional Charges:**
  Insurance Fee + Fuel Charge + Other Charges
- **Subtotal:**
  Rental Cost + Total Additional Charges

- **Discount Amount:**
  Subtotal × (Discount / 100)
- **Total Amount Due:**
  Subtotal − Discount Amount
- **Change / Remaining Balance:**
  - If Amount Paid ≥ Total Amount Due →
    Change = Amount Paid − Total Amount Due
  - If Amount Paid < Total Amount Due →
    Remaining Balance = Total Amount Due − Amount Paid

Output Requirements

Display a **well-formatted rental receipt** in the console:

******** CAR RENTAL RECEIPT ********

Customer Details

- Name
- Customer ID
- Driver's License Number

Vehicle Information

- Vehicle Type
- Vehicle Model
- Rate per Day

Rental Summary

- Pickup Date
- Return Date
- Number of Days
- Rental Cost

Charges Breakdown

- Insurance Fee
- Fuel Charge
- Other Charges
- Subtotal

Payment Summary

- Discount (%)
- Discount Amount
- Total Amount Due
- Amount Paid
- Change **or** Remaining Balance

✓ All monetary values must be formatted to **2 decimal places**
✓ Output must be **clean, aligned, and readable**
✓ Use separators for better formatting

Sample Input

- Name: Juan Dela Cruz
- ID: CUST-2026
- License: D1234567
- Vehicle Type: SUV
- Model: Toyota Fortuner
- Rate/Day: 3500
- Days: 3
- Pickup: 2026-09-01
- Return: 2026-09-04
- Insurance: 1500
- Fuel: 800

- Other Charges: 200
- Discount: 10%
- Amount Paid: 12000

PROGRAMIMING 24 **Simple ROS – Restaurant Ordering System**

You are required to develop a **Java console application** that simulates a **Restaurant Ordering System (ROS)**. The program will allow a user to input customer order details (dine-in or takeout), compute the total bill, and generate a **formatted receipt**.

This system represents a simplified restaurant process handling **orders, billing, and payments**.

Input Requirements

Use a Scanner object to collect the following:

1. Customer Information

- Customer Name
- Order Type (Dine-In or Takeout)

2. Order Details

The system must accept **at least 3 menu items**:

For each item:

- Item Name
- Quantity
- Price per Item

3. Additional Charges

- Service Charge (for Dine-In only; input 0 if Takeout)
- Packaging Fee (for Takeout only; input 0 if Dine-In)

4. Payment Details

- Discount (%)

- Amount Tendered

Computation and Business Logic

Your program must perform the following:

- **Item Total (per item):**
  Quantity × Price
- **Subtotal:**
  Sum of all item totals
- **Additional Charges:**
  - If **Dine-In** → Add Service Charge
  - If **Takeout** → Add Packaging Fee
- **Total Before Discount:**
  Subtotal + Applicable Charges
- **Discount Amount:**
  Total Before Discount × (Discount / 100)
- **Total Amount Due:**
  Total Before Discount – Discount Amount
- **Change:**
  Amount Tendered – Total Amount Due

Output Requirements

Display a **well-formatted restaurant receipt** in the console:

******** RESTAURANT RECEIPT ********

Customer Details

- Name
- Order Type

Order Summary

Display in tabular format:

- Item Name
- Quantity
- Price
- Total

Billing Summary

- Subtotal
- Service Charge **or** Packaging Fee
- Total Before Discount
- Discount (%)
- Discount Amount
- Total Amount Due
- Amount Tendered
- Change

✓ All monetary values must be formatted to **2 decimal places**
✓ Output should be **aligned and easy to read**
✓ Use separators for clean formatting

Sample Input

- Name: Maria Santos
- Order Type: Dine-In

Items:

1. Burger – Qty: 2 – Price: 120
2. Fries – Qty: 1 – Price: 80
3. Soda – Qty: 3 – Price: 50

- Service Charge: 100
- Packaging Fee: 0

- Discount: 5%
- Amount Tendered: 500

PROGRAMMING 25 **Simple Flower Reservation with Billing System**

You are required to develop a **Java console application** that simulates a **Flower Reservation and Billing System**. The program will allow a user to input customer details, reserve flower arrangements, compute the total cost, and generate a **formatted billing receipt**.

This system represents a simplified version of real-world **flower shop reservations** for events such as birthdays, weddings, and special occasions.

Input Requirements

Use a Scanner object to collect the following:

1. Customer Information

- Customer Name
- Customer ID
- Contact Number

2. Reservation Details

- Flower Type (e.g., Roses, Tulips, Sunflowers)
- Arrangement Type (e.g., Bouquet, Vase, Box)
- Price per Arrangement
- Quantity
- Reservation Date (YYYY-MM-DD)
- Pickup/Delivery Date (YYYY-MM-DD)

3. Additional Services

- Delivery Fee
- Customization Fee (e.g., ribbon, message card)
- Other Charges

4. Payment Details

- Discount (%)
- Down Payment

Computation and Business Logic

Your program must perform the following:

- **Reservation Cost:**
  Price per Arrangement × Quantity
- **Total Additional Charges:**
  Delivery Fee + Customization Fee + Other Charges
- **Subtotal:**
  Reservation Cost + Total Additional Charges
- **Discount Amount:**
  Subtotal × (Discount / 100)
- **Total Bill:**
  Subtotal – Discount Amount
- **Remaining Balance:**
  Total Bill – Down Payment

Output Requirements

Display a **well-formatted reservation receipt** in the console:

******** FLOWER RESERVATION RECEIPT ********

Customer Details

- Name
- Customer ID
- Contact Number

Reservation Summary

- Flower Type

- Arrangement Type
- Quantity
- Price per Arrangement
- Reservation Cost
- Reservation Date
- Pickup/Delivery Date

Charges Breakdown

- Delivery Fee
- Customization Fee
- Other Charges
- Subtotal

Payment Summary

- Discount (%)
- Discount Amount
- Total Bill
- Down Payment
- Remaining Balance

✓ All monetary values must be formatted to **2 decimal places**
✓ Output must be **clean, aligned, and readable**
✓ Use separators for better formatting

Sample Input

- Name: Maria Santos
- ID: FL-001
- Contact: 09123456789
- Flower Type: Roses

- Arrangement: Bouquet
- Price: 800
- Quantity: 3
- Reservation Date: 2026-09-01
- Delivery Date: 2026-09-05
- Delivery Fee: 200
- Customization Fee: 150
- Other Charges: 50
- Discount: 10%
- Down Payment: 1000

PROGRAMMING 26 **KMART Korean Noodles Purchase Order with Billing System**

You are required to develop a **Java console application** that simulates a **purchase order and billing system** for a KMART store specializing in Korean noodles. The program will collect customer details, allow ordering of noodle products, compute the total bill, and generate a **formatted receipt**.

This system represents a simplified **retail checkout system**.

Input Requirements

Use a Scanner object to gather the following:

1. Customer Information

- Customer Name
- Customer ID
- Contact Number

2. Order Details

- Product Name (e.g., Shin Ramyun, Chapagetti, Neoguri)
- Brand

- Price per Pack
- Quantity

3. Additional Charges

- Packaging Fee
- Delivery Fee (if applicable)

4. Payment Details

- Discount (%)
- Cash Payment

Computation and Business Logic

Your program must perform the following:

- **Total Product Cost:**
  Price per Pack × Quantity
- **Subtotal:**
  Total Product Cost + Packaging Fee + Delivery Fee
- **Discount Amount:**
  Subtotal × (Discount / 100)
- **Total Bill:**
  Subtotal − Discount Amount
- **Payment Calculation:**
  - If Cash Payment ≥ Total Bill →
    Change = Cash Payment − Total Bill
  - If Cash Payment < Total Bill →
    Remaining Balance = Total Bill − Cash Payment

Output Requirements

Display a **well-formatted purchase receipt**:

******** KMART PURCHASE RECEIPT ********

Customer Details

- Name
- Customer ID
- Contact Number

Order Summary

- Product Name
- Brand
- Price per Pack
- Quantity
- Total Product Cost

Charges

- Packaging Fee
- Delivery Fee
- Subtotal

Payment Summary

- Discount (%)
- Discount Amount
- Total Bill
- Cash Payment
- Change **or** Remaining Balance

✓ Format all monetary values to **2 decimal places**
✓ Ensure the output is **organized and readable**
✓ Use separators (lines) for clarity

Sample Input

- Name: Kim Lee

- ID: KM-101
- Contact: 09121234567
- Product: Shin Ramyun
- Brand: Nongshim
- Price: 75.50
- Quantity: 5
- Packaging Fee: 20
- Delivery Fee: 50
- Discount: 5%
- Cash Payment: 500

PROGRAMMING 27 **Car and Motor Parking Reservation and Billing System**

You are required to develop a **Java console application** that simulates a **Parking Reservation and Billing System** for cars and motorcycles. The program will collect customer and vehicle details, compute parking fees based on time, and generate a **formatted parking receipt**.

This system represents a simplified version of **parking management in malls, terminals, or private parking areas**.

Input Requirements

Use a Scanner object to collect the following:

1. Customer Information

- Customer Name
- Customer ID
- Contact Number

2. Vehicle Information

- Vehicle Type (Car or Motorcycle)

- Plate Number

3. Parking Reservation Details

- Entry Time (HH:MM)
- Exit Time (HH:MM)
- Hourly Rate
- Reservation Fee

4. Additional Charges

- Overnight Fee (if applicable)
- Penalty Fee (for exceeding reserved time)

5. Payment Details

- Discount (%)
- Cash Payment

Computation and Business Logic

Your program must perform the following:

- **Total Parking Hours:**
  Calculate the difference between Entry Time and Exit Time
- **Parking Fee:**
  Total Parking Hours × Hourly Rate
- **Subtotal:**
  Parking Fee + Reservation Fee + Overnight Fee + Penalty Fee
- **Discount Amount:**
  Subtotal × (Discount / 100)
- **Total Bill:**
  Subtotal − Discount Amount
- **Payment Calculation:**
  - If Cash Payment ≥ Total Bill →
    Change = Cash Payment − Total Bill

- If Cash Payment < Total Bill →
  Remaining Balance = Total Bill − Cash Payment

Output Requirements

Display a **well-formatted parking receipt**:

******** PARKING RECEIPT ********

Customer Details

- Name
- Customer ID
- Contact Number

Vehicle Details

- Vehicle Type
- Plate Number

Parking Summary

- Entry Time
- Exit Time
- Total Hours
- Hourly Rate
- Parking Fee

Charges Breakdown

- Reservation Fee
- Overnight Fee
- Penalty Fee
- Subtotal

Payment Summary

- Discount (%)

- Discount Amount
- Total Bill
- Cash Payment
- Change **or** Remaining Balance

✓ All monetary values must be formatted to **2 decimal places**
✓ Time calculations should be handled correctly
✓ Output must be **clean, aligned, and readable**
✓ Use separators for better presentation

Sample Input

- Name: Alex Cruz
- ID: PK-202
- Contact: 09123456789
- Vehicle Type: Car
- Plate Number: ABC-1234
- Entry Time: 08:00
- Exit Time: 12:30
- Hourly Rate: 50
- Reservation Fee: 20
- Overnight Fee: 0
- Penalty Fee: 0
- Discount: 10%
- Cash Payment: 300